

Chess Thermodynamics

A First-Principles Evaluation Function via Statistical Mechanics of the Entropy Field

Technical Specification — June 2026

This document specifies a chess evaluation function derived from statistical mechanics. The board is modeled as a scalar entropy field on a 64-site lattice. Each piece generates a charge equal to the logarithm of its mobility. The evaluation is the entropy difference between sides, corrected by empirically evolved interaction terms expressed as a multipole expansion of the field in the position's intrinsic coordinate system. The framework separates cleanly into an axiomatic layer (no free parameters, fully determined by the rules of chess and the uniqueness of logarithmic entropy) and an evolvable layer (interaction coefficients discovered by self-play). This document is intended as a complete implementation specification.

1. Architecture Overview

The system has two layers with a hard boundary between them. The axiomatic layer takes a chess position and returns a fixed-dimensional feature vector. It has zero free parameters. The evolvable layer takes the feature vector and a weight vector and returns a scalar evaluation in centipawns. All learnable content lives in the weight vector. All principled content lives in the feature computation.

The evaluation function is:

$$E(P) = w \cdot f(P) + w_2 \cdot (f(P) \otimes f(P))$$

where $f(P)$ is the axiomatic feature vector, w is the linear weight vector, w_2 is the quadratic interaction matrix (flattened), and $\otimes$ denotes the outer product. The linear term captures single-moment effects. The quadratic term captures pairwise interactions between spatial modes of the entropy field — the virial B_2 coefficients.

2. The Axiomatic Layer

Everything in this section is determined by the rules of chess, the uniqueness theorem for logarithmic entropy, and standard linear algebra. There are no design choices and no free parameters.

2.1 The Entropy Charge

Axiom 1 (Entropy Charge).

For a piece at square s with mobility m_s (the number of squares it attacks), the entropy charge is $q_s = \ln(m_s)$. If $m_s = 0$, set $q_s = -C$ for a fixed penalty constant $C > 0$ (a frozen piece contributes negative entropy).

The logarithm is not a design choice. It is the unique function satisfying three requirements: (i) additivity under independent composition (if two non-interacting pieces have mobilities m_1 and m_2 , their joint multiplicity is $m_1 \cdot m_2$, and the joint entropy must equal the sum of individual entropies); (ii) monotonicity (more mobile pieces have higher entropy); (iii) continuity. This is the same uniqueness argument that establishes $S = k \ln \Omega$ in statistical mechanics.

2.2 The Entropy Field

Axiom 2 (Entropy Field).

The entropy field is a scalar field on the 8×8 lattice defined by: $\phi(s) = +q_s$ if a White piece occupies s ; $\phi(s) = -q_s$ if a Black piece occupies s ; $\phi(s) = 0$ if s is empty.

The sign convention is forced by the color antisymmetry requirement: flipping all colors and mirroring the board must negate the evaluation. This means E must be a functional of the signed field ϕ , not of the unsigned charges.

Piece classification. Pieces are divided into two mechanical categories:

- **Thermal particles** (queens, rooks, bishops, knights): fully mobile, can be rearranged across the board. These generate the primary entropy charges. Their contribution to ϕ is weighted by 1.0.
- **Structural elements** (pawns and kings): pawns are irreversible (forward-only) and define the lattice geometry. Kings are permanent and define the boundary conditions. Their contributions to ϕ are weighted by learned scalars w_{pawn} and w_{king} (the only two parameters that cross the layer boundary; see Section 3).

2.3 Thermodynamic State Variables

Definition (Volume).

$V = 62 - n_p$, where n_p is the total number of pawns on the board. V counts the squares available for thermal particles (64 minus 2 kings minus n_p pawns). Pawns are the walls of the container.

Definition (Particle Number).

$N = N^w + N^b$, the total count of thermal particles (both colors). N^w and N^b are the per-side counts.

These are observable properties of the position, not parameters. The chemical potential of a pawn is identically equal to the pressure, because adding a pawn simultaneously removes an available square:

$\mu_{\text{pawn}} = -T(\partial S/\partial V)(\partial V/\partial n_p) = T(\partial S/\partial V) = P$. Pawns are both matter and structure; their dual nature is what gives PV a well-defined meaning.

2.4 Intrinsic Coordinate System

The standard file–rank coordinate system is arbitrary — it reflects the labeling of the board, not the strategic structure of the position. The entropy field defines its own natural coordinate system, which adapts to each position.

Definition (Center of Entropy).

The center of entropy is the charge-weighted centroid of the field: $\mathbf{r}_{\blacksquare} = \sum_s |\phi(s)| \cdot \mathbf{r}(s) / \sum_s |\phi(s)|$, where $\mathbf{r}(s) = (\text{file}(s), \text{rank}(s))$ is the coordinate of square s .

Definition (Inertia Tensor).

The inertia tensor of the entropy field is the 2×2 matrix: $I_{ij} = \sum_s |\phi(s)| \cdot \Delta r_i(s) \cdot \Delta r_j(s)$, where $\Delta \mathbf{r}(s) = \mathbf{r}(s) - \mathbf{r}_{\blacksquare}$ is the displacement from the center of entropy.

Definition (Principal Axes).

Diagonalize I to obtain eigenvalues $\lambda_1 \geq \lambda_2$ and eigenvectors $\mathbf{e}_{\blacksquare_1}, \mathbf{e}_{\blacksquare_2}$. These are the position's intrinsic axes. $\mathbf{e}_{\blacksquare_1}$ is the major axis (direction of maximum entropy spread); $\mathbf{e}_{\blacksquare_2}$ is the minor axis. All subsequent multipole moments are computed in this eigenbasis.

This construction is a pure function of the entropy field. It has no free parameters, no reference to game history, and no dependence on board orientation. Two positions that are strategically equivalent (related by rotation/reflection of the board) will produce the same intrinsic axes up to the same transformation. The principal axes capture the position's own geometry: which direction is the main line of engagement, and which is peripheral.

2.5 Multipole Expansion in the Eigenbasis

Express the entropy field in the intrinsic coordinate system (u_1, u_2) obtained by projecting each square's displacement onto the principal axes. Then compute moments of the field. These moments are the feature vector $\mathbf{f}(P)$.

The features are organized by order and by color decomposition:

Monopole (0th order)

- $S^w = \sum_{s \in \text{White}} q_s$ — total white entropy
- $S^b = \sum_{s \in \text{Black}} q_s$ — total black entropy
- $\Delta S = S^w - S^b$ — entropy difference (the ideal gas evaluation, before scaling)

Dipole (1st order)

- D_1^w, D_1^b — projection of each color's entropy onto the major axis
- D_2^w, D_2^b — projection onto the minor axis

The dipole moments measure where entropy is concentrated along the position's natural axes. A large D_1^w means white's mobile pieces are displaced along the major axis from the center of the entropy distribution. This captures concepts like flank concentration and advancement without naming any specific direction.

Quadrupole (2nd order)

- Q_{11}^w, Q_{11}^b — entropy spread along major axis
- Q_{22}^w, Q_{22}^b — entropy spread along minor axis
- Q_{12}^w, Q_{12}^b — cross-correlation between axes

The quadrupole moments measure the shape of the entropy distribution. A position where one side's entropy is spread along the major axis but narrow along the minor axis has very different Q_{11}/Q_{22} ratios than a position with isotropically spread entropy. This captures positional concepts like overextension and spatial control.

King-Field Coupling

The king is a boundary condition, not a thermal particle. Its effect enters through the coupling between the entropy field and the king's position. Define a radial kernel around each king: $K(s, \text{king}) = 1/(1 + d(s, \text{king}))$, where d is the Chebyshev distance. Then:

- $A_{wk} = \sum_s \phi^b(s) \cdot K(s, wk)$ — black entropy near white king (attack pressure)
- $D_{wk} = \sum_s \phi^w(s) \cdot K(s, wk)$ — white entropy near white king (defense)
- $A_{bk} = \sum_s \phi^w(s) \cdot K(s, bk)$ — white entropy near black king (attack pressure)
- $D_{bk} = \sum_s \phi^b(s) \cdot K(s, bk)$ — black entropy near black king (defense)

These are the boundary terms. They contain no information about piece types — only about how much entropy density exists in the king's neighborhood.

Meta-Entropy and Concentration

- $\eta^w = -\sum p_i \ln(p_i)$ where $p_i = q_i/S^w$ — dispersion of white entropy across its pieces
- η^b — same for black

The meta-entropy measures whether one side's entropy is concentrated in a few powerful pieces or spread uniformly across many. Concentrated entropy (low η) means a small number of pieces are doing all the work — a tactical, piece-dependent position. Dispersed entropy (high η) means many pieces contribute equally — a positional, structure-dependent position.

Pawn Lattice Features

The pawn structure defines the container geometry. These features describe the container's shape, distinct from its contents:

- Total pawn count per side
- Pawn advancement (sum of rank distances from starting rank)
- Pawn lateral spread (variance of pawn file positions)

Specific pawn patterns (passed, isolated, doubled, connected) could be included as additional lattice features. Their importance is an empirical question answered by the evolution.

Spatial Tension and Temperature

- $\tau = \sum_s \phi^w(s) \cdot \phi^b(s)$ — spatial overlap of white and black entropy fields
- $T_{\blacksquare} = V/N$ — the ideal gas temperature (included as a feature so the evolvable layer can learn T-dependent interactions)

Spatial tension τ is nonzero when opposing pieces are on the same or adjacent squares. It measures the degree to which the two sides are in contact, distinguishing engaged positions from separated ones.

The complete feature vector $f(P)$ thus has approximately 25–30 components (the exact count depends on which pawn structure features are included). All components are computed from the rules of chess, the entropy charges, and standard linear algebra. None depend on piece names, opening theory, or game history.

3. The Evolvable Layer

This section describes everything that is learned from self-play. The boundary between axiomatic and evolvable is sharp: the feature vector $f(P)$ is fixed; only the mapping from $f(P)$ to the scalar evaluation E is subject to evolution.

3.1 Parameters

The complete set of evolvable parameters:

- $w_{\text{pawn}} \in [0, 1]$: the weight of pawn mobility in the entropy field. At $w_{\text{pawn}} = 0$, pawns are pure structure (they define V but contribute no charge). At $w_{\text{pawn}} = 1$, pawns are fully thermal (same status as pieces). Initial guess: 0.5.
- $w_{\text{king}} \in [0, 1]$: the weight of king mobility in the entropy field. Initial guess: 0.3.
- $w_{\text{linear}} \in \mathbb{R}^n$: one weight per multipole feature. These capture single-moment effects (e.g., 'entropy advancement is good' or 'high dispersion is bad'). Dimension: $n \approx 27$.
- $W_{\text{quad}} \in \mathbb{R}^{n \times n}$ (symmetric): one weight per pair of multipole features. These are the B_2 virial coefficients — the interaction strengths between spatial modes of the entropy field. Dimension: $n(n+1)/2 \approx 378$. Most should be zero (sparse).

Total parameter count: approximately 400. Compare to NNUE (~40 million) and handcrafted evaluation functions (~200 hand-tuned terms). This framework sits between them in expressiveness, and exceeds both in interpretability.

3.2 The Evaluation Function

The full evaluation, assembled from both layers:

$$E(P) = w_{\text{linear}} \cdot f(P) + \sum_{i \leq j} W_{ij} \cdot f_i(P) \cdot f_j(P)$$

The ideal gas term is not hardcoded separately. It is subsumed into the linear term: the feature $\Delta S = S^w - S^b$ is present in $f(P)$, and its weight in w_{linear} plays the role of $\lambda \cdot T$ in the original formulation. If the ideal gas is a good zeroth-order approximation, evolution will assign a large positive weight to ΔS . If it is not, evolution will distribute weight across other features.

The quadratic terms capture the virial corrections. The product $f_i \cdot f_j$ represents the interaction between spatial modes i and j . For example, $W(A_{\text{bk}}, T_0) > 0$ would mean that king attacks become more valuable at higher temperature (more valuable when pieces are sparse). $W(D_1^w, D_1^b) < 0$ would mean that concentrating both sides' entropy along the same axis is mutually costly (contested flanks are zero-sum).

3.3 Sparsity

The quadratic interaction matrix should be sparse. In real chemical solutions, most binary interaction coefficients are negligible — most component pairs don't interact strongly. The same is expected in chess: most pairs of spatial entropy modes have no meaningful coupling. Only a few specific interactions

(king attack $\times$ temperature, entropy dipole $\times$ entropy dipole, etc.) should carry weight.

Sparsity is enforced during evolution by an L1 penalty on the quadratic weights: $\text{fitness}(\text{individual}) = \text{tournament_score} - \alpha \cdot \sum |W_{ij}|$. This encourages the search to concentrate weight in a small number of important interactions rather than spreading small weights across many. The penalty coefficient α controls the tradeoff between playing strength and parsimony.

4. Thermodynamic Foundations

4.1 The Equilibrium Hypothesis

Under optimal play by both sides, the evaluation is approximately conserved: $\Delta E \approx 0$ at every ply. White maximizes E , Black minimizes E , and these forces balance in equilibrium. The game evolves along a constant- E surface in (V, N) space.

This is a constraint on the evaluation function, not an assumption about it. The evaluation is defined as the unique state function that is conserved under optimal play. The functional form $E = w \cdot f + \text{quadratic terms}$ is the parameterized family within which evolution searches for the most nearly conserved quantity.

A perfect evaluation function would be exactly conserved under perfect play. The gap between the evolved E and perfect conservation measures the sum of all missing interaction terms. Improving the evaluation is equivalent to reducing this gap.

4.2 Derived Thermodynamic Quantities

Once E is defined, the thermodynamic potentials follow from the density of states. For a given macrostate (fixed V, N , pawn structure, king positions), count the number of legal piece arrangements Ω that produce evaluation E . Define $S = \ln \Omega$. Then:

$$1/T = \partial S / \partial E \big|_{V, N}$$

$$P/T = \partial S / \partial V \big|_{E, N}$$

$$\mu_i / T = -\partial S / \partial N_i \big|_{E, V}$$

$$F = E - TS \text{ (Helmholtz free energy)}$$

$$H = E + PV \text{ (Enthalpy)}$$

$$G = H - TS \text{ (Gibbs free energy)}$$

In the ideal gas limit: $PV = NT$, giving $T = V/N$. This is a lower bound on the true temperature. The correction factor depends on the heterogeneity of the piece-square landscape — how much the value of a square varies across the board. In practice, T should be measured by Monte Carlo sampling (permuting

pieces within the macrostate, evaluating each arrangement, and computing the standard deviation of E) rather than assumed from V/N .

4.3 The Pawn Identity

Because $V = 62 - n_p$, adding a pawn decreases V by 1. The chemical potential of a pawn is therefore:

$$\mu_{\text{pawn}} = -T(\partial S/\partial V)(\partial V/\partial n_p) = T(\partial S/\partial V) = P$$

This is an exact identity, not an approximation. It connects the pawn's material value to the pressure of the thermal gas of pieces. In the ideal gas limit, $\mu_{\text{pawn}} = P = NT/V = T$, so a pawn is worth exactly one temperature unit. Since T decreases as pieces are traded (V/N increases but N decreases faster), the pawn's value in temperature units increases in the endgame — the well-known principle that pawns become more valuable as pieces come off.

5. Evolutionary Training Protocol

5.1 Fitness Signal

The only non-circular fitness signal is game outcome. Using an existing engine's evaluation as the target would train the system to approximate that engine, not to discover the true interaction structure. Self-play with game outcomes (win = +1, draw = 0, loss = -1) is the correct signal.

A won game means the evaluation function (with these particular weights) consistently identified moves that led to a terminal advantage. A lost game means it did not. The weights that produce the most wins encode the interaction structure that matters for actual chess performance.

5.2 Population Dynamics

The evolutionary algorithm:

- **Initialization:** A population of K individuals, each carrying a weight vector. One individual is pure ideal gas (all weights zero). One is seeded with rough priors. The rest are random with sparse quadratic terms (~10% nonzero).
- **Tournament:** Round-robin self-play. Each pair plays G games, alternating colors. Fitness = total score minus L1 sparsity penalty on quadratic weights.
- **Selection:** Top fraction survives unchanged (elitism). Bottom fraction is replaced.
- **Reproduction:** Children are produced by crossover (gene-level: each weight is independently inherited from one parent) followed by mutation (Gaussian perturbation of a random subset of weights).
- **Sparsity maintenance:** Quadratic weights below a threshold are zeroed. New quadratic terms are occasionally activated at random (exploration). This encourages the population to discover which spatial mode couplings matter, not just to tune all of them slightly.
- **Annealing:** Mutation scale decreases over generations, shifting from exploration to exploitation.

5.3 Search During Self-Play

Each individual plays using alpha-beta search at fixed depth (depth 1–2 during training for speed) with quiescence extension on captures (2–3 additional plies for capture-only lines). Move ordering: checks first, then captures sorted by victim value, then quiet moves. The evaluation function at leaf nodes uses that individual's weight vector.

Training games are capped at 50–100 moves. Positions with $|E| > 5000$ cp are adjudicated as decisive. This prevents wasting computation on games that are clearly won or lost but not yet technically over.

5.4 Scaling Considerations

With population K and G games per pair, each generation requires $K(K-1)G/2$ games. At depth 1 with quiescence, each game takes 3–5 seconds in Python. Recommended minimum: $K = 20$, $G = 2$, 50 generations = 19,000 games $\approx$ 20 hours on a single core. For publishable results: $K = 50$, $G = 4$, 200 generations $\approx$ two weeks on a single core, or a day on a 16-core machine. A C++ implementation would reduce per-game time by 50–100 $\times$.

6. Physical Interpretation

6.1 What the Interactions Mean

Each nonzero W_{ij} in the quadratic interaction matrix has a physical interpretation as a coupling between two spatial properties of the entropy field. These are the virial B_2 coefficients of the chess equation of state. Representative examples from preliminary evolution:

- $W(\Delta S, T_0) > 0$: the entropy difference matters more when temperature is high. Material/mobility advantages are amplified when pieces are sparse. This is the endgame principle quantified.
- $W(A_{bk}, T_0) > 0$: king attacks are more valuable at higher temperature. Each attacking piece near the king contributes more when there are fewer total pieces (less defensive coverage).
- $W(D_1^w, D_1^b) < 0$: when both sides concentrate entropy along the same axis, the interaction is negative. This captures the zero-sum nature of a contested region.
- $W(\eta^w, \eta^b) < 0$: when both sides have dispersed entropy, the interaction is costly. Concentrated vs. dispersed has an advantage; mutual dispersion is mutual disadvantage.

6.2 The Equation of State for Chess

The full evaluation function, once trained, constitutes an equation of state for chess: a relationship between the thermodynamic variables (S, V, N, T, P, μ) and the internal energy E . In physics, the equation of state of a system (e.g., $PV = NkT$ for an ideal gas, or the van der Waals equation for interacting particles) captures the macroscopic behavior of the system without resolving individual particle dynamics.

Here, the equation of state captures the macroscopic behavior of chess positions without resolving individual move sequences. It tells you the value of a position from its thermodynamic profile — its entropy distribution, temperature, and interaction structure — without searching the game tree. The search (alpha-beta) is then a refinement: it resolves the microstate dynamics (specific move sequences) that the equation of state cannot capture.

The interaction terms W_{ij} are the chess-specific deviations from ideality. They encode the information that an ideal gas of non-interacting pieces misses: coordination, king safety, structural weaknesses, and tactical motifs. Each W_{ij} is analogous to a Margules parameter in solution thermodynamics or a Lennard-Jones coefficient in molecular simulation — an empirically measured quantity that characterizes the interaction between two components of the system.

7. Summary of Design Decisions

The framework rests on the following chain of reasoning:

Step	Content	Status
1	Mobility $m(s)$ is the natural state variable for a piece on square s .	Axiomatic (from rules)
2	$\ln(m)$ is the unique additive, monotone, continuous entropy measure.	Axiomatic (uniqueness)
3	The signed entropy field $\phi(s)$ on the 8×8 lattice encodes the full position in thermodynamic terms.	Axiomatic (color symmetry)
4	The position's intrinsic axes come from the eigendecomposition of the field's inertia tensor.	Axiomatic (linear algebra)
5	The multipole moments in the eigenbasis are the feature vector.	Axiomatic (standard expansion)
6	The evaluation is a linear + quadratic function of the features.	Structural choice
7	The weights are discovered by evolutionary self-play.	Empirical (evolved)
8	Sparsity of the quadratic matrix is enforced by L1 penalty.	Structural choice

Steps 1–5 have zero free parameters. Steps 6 and 8 are structural choices that could be revised (e.g., higher-order terms, different regularization). Step 7 is the empirical content — the only place where data enters the system. The resulting evaluation function is interpretable (each weight has a named physical meaning), compact (~400 parameters), and derived from first principles up to the interaction corrections, which are measured from self-play in the same way that virial coefficients are measured from experiment.

End of specification.